Kernelized polynomial regression

Formula overview

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Notation follows the lecture. $\mathbb{N} = \{0, 1, 2, \ldots\}$. A multi-index $\alpha = (\alpha_1, \ldots, \alpha_d) \in \mathbb{N}^d$ has $x^\alpha = \prod_{i \le d} x_i^{\alpha_i}$ and $\alpha! = \prod_{i \le d} \alpha_i!$.

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1 Risk

Unknown regression function $f^*\colon \mathbb{R}^d \to \mathbb{R}$. Training pairs (x_i, y_i), $i \le n$, drawn from p^*, with

$$y_i = f^*(x_i) + \varepsilon_i^*, \qquad \varepsilon_i^* \sim \mathcal{N}(0, \sigma^2). \tag{1}$$

The same model is used for a future sample (x_i^*, y_i^*), $i \le m$.

The quantity to minimize on future data, and its expectation, are

$$\hat{R}_*(f) = \frac{1}{m} \sum_{i \le m} \big(y_i^* - f(x_i^*)\big)^2, \tag{2}$$

$$R(f) = \mathbb{E}_{(X,Y)\sim p^*}\big[(Y - f(X))^2\big]. \tag{3}$$

Under (1) the expected risk splits into approximation and noise,

$$R(f) = \mathbb{E}\big[(f(X) - f^*(X))^2\big] + \sigma^2. \tag{4}$$

The training substitute is the empirical risk

$$\hat{R}(f) = \frac{1}{n} \sum_{i \le n} \big(y_i - f(x_i)\big)^2. \tag{5}$$

2 Least squares and ridge

Two routes give the same linear estimator. Maximum likelihood with $y_i \sim \mathcal{N}(\theta^\top x_i, \sigma^2)$ is

$$\hat{\theta} = \arg\min_{\theta \in \mathbb{R}^d} \sum_{i \le n} \big(y_i - \theta^\top x_i\big)^2, \tag{6}$$

because the Gaussian likelihood is a decreasing function of that sum.

Statistical learning restricts $\hat{R}$ to a hypothesis class and drops the factor $1/n$, which does not change the minimizer. The linear class and the ordinary least-squares estimator (OLSE) are

$$\mathcal{H}_l = \big\{ f \colon \mathbb{R}^d \to \mathbb{R} \mid f(x) = w^\top x \text{ for some } w \in \mathbb{R}^d \big\}, \tag{7}$$

$$\hat{w} = \arg\min_{w \in \mathbb{R}^d} \sum_{i \le n} \big(y_i - w^\top x_i\big)^2 = (X^\top X)^{-1} X^\top y, \tag{8}$$

with design matrix and response

$$X = \begin{bmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{bmatrix} \in \mathbb{R}^{n \times d}, \qquad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n. \tag{9}$$

So $\hat{\theta}_{\mathrm{MLE}} = \hat{w}_{\mathrm{OLSE}}$. Building $X^\top X$, building $X^\top y$, and solving the $d \times d$ system cost $O(nd^2 + d^3)$.

If $y_i = x_i^\top w^* + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$, the OLSE is unbiased,

$$\mathbb{E}_{y|X}[\hat{w}] = w^*. \tag{10}$$

Among unbiased estimators of the form $\tilde{w} = Cy$, it has the smallest variance at every test point (Gauss-Markov):

$$\mathrm{Var}\big(x_*^\top \hat{w}\big) \le \mathrm{Var}\big(x_*^\top \tilde{w}\big). \tag{11}$$

Pointwise error still splits as

$$\mathbb{E}\big[(y_* - \hat{f}(x_*))^2\big] = \big(\mathbb{E}[\hat{f}(x_*)] - f^*(x_*)\big)^2 + \mathrm{Var}\big(\hat{f}(x_*)\big) + \sigma^2. \tag{12}$$

The OLSE term in the middle is often large. Ridge pays a little bias to shrink it,

$$\hat{R}_r(w) = \|y - Xw\|^2 + \lambda \|w\|^2, \tag{13}$$

$$\hat{w}_r = \arg\min_w \hat{R}_r(w) = (X^\top X + \lambda I)^{-1} X^\top y. \tag{14}$$

3 Polynomial features

$\mathcal{H}_l$ cannot approximate a general continuous target. On any closed bounded $B \subset \mathbb{R}^d$, polynomials can: for every continuous $g \colon B \to \mathbb{R}$ and every $\delta > 0$ there is a polynomial f with $|f(x) - g(x)| < \delta$ for all $x \in B$ [5, 4]. Write $\mathcal{H}_p$ for that class.

Every $f \in \mathcal{H}_p$ is linear in a monomial coordinate. One variable, every degree:

$$f(x) = \sum_{k \in \mathbb{N}} w_k x^k = \langle w, \tilde{\varphi}(x) \rangle, \qquad \tilde{\varphi}(x) = (1, x, x^2, x^3, \ldots). \tag{15}$$

In d variables the same statement uses one index per coordinate,

$$f(x) = \sum_{\alpha \in \mathbb{N}^d} w_\alpha x^\alpha = \langle w, \tilde{\varphi}(x) \rangle. \tag{16}$$

Raw monomials are unusable as features: at $x = 1$ and $w = (1, 1, \ldots)$ the inner product diverges, and for $d = 100$ the monomials of degree 5 already number almost 10^8.

Scale each monomial by a Gaussian bump and a factorial,

$$\varphi(x) = e^{-\|x\|^2/2}\left[\frac{x^\alpha}{\sqrt{\alpha!}}\right]_{\alpha\in\mathbb{N}^d}. \tag{17}$$

Then $\|\varphi(x)\| = 1$ for every x, there is no degree cutoff, and the model remains a universal approximator. The fit is still linear regression on features,

$$f(x) = \langle w, \varphi(x)\rangle. \tag{18}$$

The one-variable map is the same formula with $\|x\|^2 = x^2$ and $\alpha = k$.

4 The kernel trick

Stack the feature rows $\Phi \in \mathbb{R}^{n\times\infty}$, with row i equal to $\varphi(x_i)^\top$. Polynomial least squares asks for

$$\hat{w}_\infty = \operatorname*{arg\,min}_{w\in\mathbb{R}^\infty} \|\Phi w - y\|^2. \tag{19}$$

The normal equation $(\Phi^\top\Phi)^{-1}\Phi^\top y$ is an infinite matrix. The minimum-norm solution never forms it,

$$\hat{w}_\infty = \Phi^\top(\Phi\Phi^\top)^{-1}y. \tag{20}$$

(The lecture treats the usual normal equation as shorthand: $\Phi^\top\Phi$ has no inverse. Equation (20) is the exact minimum-norm solution.)

Prediction at $x_* \in \mathbb{R}^d$ uses only $n \times n$ kernel entries. The sum over all scaled monomials collapses to the Gaussian kernel,

$$k(x, x') = \langle\varphi(x), \varphi(x')\rangle = \exp\Big(-\frac{1}{2}\|x - x'\|^2\Big). \tag{21}$$

With $K_{ij} = k(x_i, x_j)$ and $k_{*,i} = k(x_*, x_i)$,

$$\hat{f}(x_*) = \varphi(x_*)^\top\Phi^\top(\Phi\Phi^\top)^{-1}y = k_*^\top K^{-1}y. \tag{22}$$

The same identity with a ridge penalty is exact, not a minimum-norm convention,

$$\hat{w}_\infty = \operatorname*{arg\,min}_{w\in\mathbb{R}^\infty}\Big(\|\Phi w - y\|^2 + \lambda\|w\|^2\Big) = \Phi^\top(\Phi\Phi^\top + \lambda I)^{-1}y, \tag{23}$$

$$\hat{f}(x_*) = k_*^\top(K + \lambda I)^{-1}y. \tag{24}$$

Algorithm: build K from the training points, solve $(K + \lambda I)\beta = y$, then for each test point build k_* and return $k_*^\top\beta$. Cost $O(n^2d + n^3)$. At $n = 10^6$ that is about 10^{18} operations and on the order of 8 TB for K.

5 Random features

Keep the predictor, replace the infinite map by $2Q$ numbers with $2Q < n$. Draw frequencies $\omega_q \sim \mathcal{N}(0, I_d)$ once, before seeing data, and set

$$\psi(x) = \frac{1}{\sqrt{Q}}\begin{pmatrix}\cos(\omega_1^\top x)\\ \sin(\omega_1^\top x)\\ \vdots\\ \cos(\omega_Q^\top x)\\ \sin(\omega_Q^\top x)\end{pmatrix} \in \mathbb{R}^{2Q}. \tag{25}$$

Rows $\psi(x_i)^\top$ form $\Psi \in \mathbb{R}^{n \times 2Q}$. The estimator is ordinary ridge regression in these coordinates,

$$\hat{f}(x_*) = \psi(x_*)^\top (\Psi^\top \Psi + \lambda I)^{-1} \Psi^\top y. \tag{26}$$

Cost $O(nQd + nQ^2 + Q^3)$, linear in the sample size.

It matches kernel ridge when $\psi(x)^\top \psi(x')$ matches $k(x, x')$. Bochner's theorem writes the Gaussian kernel as a Fourier expectation [1, 3]. Rahimi and Recht turn that expectation into features [2]. Because k is real and even, the expectation is an average of cosines,

$$k(x, x') = \kappa(x - x') = \mathbb{E}_{\omega \sim \mathcal{N}(0, I_d)} \big[\cos\big(\omega^\top (x - x')\big)\big]. \tag{27}$$

The average of Q independent draws is

$$k(x, x') \approx \frac{1}{Q} \sum_{q \le Q} \cos\big(\omega_q^\top (x - x')\big) = \psi(x)^\top \psi(x'). \tag{28}$$

The cosine splits by the angle formula, which is why ψ stores both $\cos(\omega_q^\top x)$ and $\sin(\omega_q^\top x)$. The weak law puts the error near $1/\sqrt{2Q}$. At $Q = 5000$ that is about 0.01.

Then $K \approx \Psi \Psi^\top$ and $k_* \approx \Psi \psi(x_*)$. The push-through identity

$$\Psi^\top (\Psi \Psi^\top + \lambda I)^{-1} = (\Psi^\top \Psi + \lambda I)^{-1} \Psi^\top \tag{29}$$

turns the $n \times n$ kernel predictor into the $2Q \times 2Q$ estimator (26).

Table 1: Three estimators of the same squared loss.

Estimator	Coordinates	Work
Linear ridge	$x \in \mathbb{R}^d$	$O(nd^2 + d^3)$
Kernel ridge	$\varphi(x) \in \mathbb{R}^\infty$	$O(n^2 d + n^3)$
Random features	$\psi(x) \in \mathbb{R}^{2Q}$	$O(nQd + nQ^2 + Q^3)$

The kernel buys every scaled monomial. Random features buy the same predictor back at a cost linear in n.

6 Source index

Formulas in this note are transcribed from the lecture slides. Named theorems are the ones the slides cite. Paths below are files in this repository.

Table 2: Internal sources.

Role	Path
Lecture slides	`experiments/006/kernelized-polynomial-regression-slides.pdf`
This note	`experiments/006/formulas/main.tex`

References

[1] Salomon Bochner. Monotone Funktionen, Stieltjessche Integrale und harmonische Analyse. *Mathematische Annalen*, 108:378–410, 1933.

[2] Ali Rahimi and Benjamin Recht. Random features for large-scale kernel machines. In *Advances in Neural Information Processing Systems*, 2007.

[3] Walter Rudin. *Fourier Analysis on Groups*. Interscience, 1962.

[4] Marshall H. Stone. The generalized Weierstrass approximation theorem. *Mathematics Magazine*, 21(4):167–184, 1948.

[5] Karl Weierstrass. Über die analytische darstellbarkeit sogenannter willkürlicher Functionen einer reellen Veränderlichen. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften zu Berlin*, pages 633–639, 1885.